

HENRIQUE FERROLHO

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⚙️ SKILLS

Leadership and Teamwork

- Competent in leading technical projects, including planning, execution, and reporting;
- Experienced in leading and working in interdisciplinary teams, fostering collaboration and innovation;
- Experienced in mentoring junior engineers and interns, providing technical guidance and support;
- Capable of interviewing and onboarding new team members, contributing to the hiring process.

Robotics and Autonomous Systems

- Proficient in motion planning, including trajectory optimization;
- Skilled in numerical optimization, whole-body control, and rigid-body dynamics;
- Hands-on experience in working with complex and high-dimensional robots, such as:
NASA Valkyrie (humanoid) · TALOS (humanoid) · ANYmal B (quadruped) · Kinova Jaco (arm) · UR10e (arm)
- Experienced in using *Robot Operating System* (ROS 1 and 2), MoveIt, SMACH, and Vicon (motion capture).
- Intermediate skills in CAD design with Onshape and 3D printing with BambuStudio and Bambu Lab printers.

Software Engineering

- Strong foundation in computer science, including:
algorithms and data structures · programming paradigms (object-oriented, functional, and logic) · databases
operating systems · networking · compilers · computer graphics · web development (back-end and front-end)
- Programming languages: C/C++, Julia, Python. Operating systems: Linux, macOS, Windows;
- Competent with optimisation and machine learning libraries: Ipopt, Knitro, PyTorch;
- Experienced in Bash scripting, CI/CD, Docker, gRPC, Protocol Buffers (protobuf);
- Familiar with agile processes (Scrum, Kanban), and Atlassian tools (Confluence, Jira, Trello).

Communication

- Very good at preparing presentations with graphics, animations, and diagrams; (e.g., 📺 youtu.be/pV4s7hzUgjc)
- Capable of communicating complex ideas in a clear and concise manner, catering to different audiences;
- Experienced in producing high-quality manuscripts, including scientific publications and reports;
- Competent in L^AT_EX, GIMP, Kdenlive, PowerPoint, and Audacity.

🔧 WORKING EXPERIENCE

ALL3 London, UK

📺 youtu.be/bATpj4PFvDQ

Technical Lead, Robotics

Sep 2025 – Present

Senior Engineer, Robotics

Aug 2024 – Aug 2025

- Drove cross-functional integration efforts across mechanical, electrical, and software teams.
- Coordinated high-profile demos and investor-facing milestones, including real-world testing and troubleshooting.
- Lead contributor to the development and testing of advanced locomotion and manipulation capabilities.
- Delivered multiple core features across locomotion, manipulation, teleoperation, optimisation, and control pipelines.
- Collaborated with teams across UK and EU offices to align development and testing workflows.
- Recognised internally for leadership potential and entrusted with responsibility for key strategic initiatives.

Ocado Technology London, UK

Robotic manipulation for picking groceries from totes using a parallel jaw gripper — robust to varying light conditions, vibrations from warehouse operations, and object diversity (various shapes, sizes, weights, friction, and fragilities).

Senior Robotics Engineer Advanced Technology Dec 2023 – Jul 2024

Robotics Engineer Advanced Technology Apr 2022 – Nov 2023

Developed and helped design a motion planning and control platform used across different teams, products, and robots — and migrated one of the robotics products from a custom (legacy) system to this new platform. Developed, tested, and integrated novel parallel jaw grippers. Implemented prototypes for slippage detection and grasp stability analysis through tactile sensing, as well as grasp state classification through visual sensing using machine learning. Contributed to the transition from retro-fitted picking stations to on-grid robot picking (OGRP).

The University of Edinburgh Edinburgh, UK

Nov 2021 – Mar 2022

Research Associate Statistical Machine Learning and Motor Control (SLMC) Group

Led the research and development of the motion planning framework for *ANYmal Bull* (a quadruped robot mounted with an arm), as part of the research project for Offshore Energy Asset Integrity Management at the UK Robotics and Artificial Intelligence Hub — orcahub.org. Created a remote inspection and asset interaction solution tailored for industrial sites.

University of Porto, Porto, Portugal

Jul – Sep, 2016

Research Software Engineer DokuWeaki

Developed a collaborative real-time editor *plugin* for DokuWiki. Maintained and improved existing DokuWiki plugins. DokuWeaki is a bundle of plugins for DokuWiki especially targeted for agile teams, supporting agile documentation.

University of Porto, Porto, Portugal

Jul – Sep, 2015

Full-Stack Developer Nutriscience

Designed and implemented a responsive online platform with *Laravel*, which was used by ~1500 households. Liaised with a team of designers, nutrition specialists, and multimedia creators. Nutriscience is an award-winning European project whose goal was to improve peoples' nutrition and eating habits.

EDUCATION

The University of Edinburgh, Edinburgh, UK

Sep 2017 – Mar 2022

PhD in Robotics and Autonomous Systems

The University of Edinburgh, Edinburgh, UK

Sep 2016 – May 2017

Erasmus Exchange Programme

Courses: Bioinformatics, Extreme Computing, Robotics, Social and Technological Networks

University of Porto, Porto, Portugal

Sep 2012 – Jul 2017

BSc+MSc in Informatics and Computing Engineering

TEACHING EXPERIENCE

The University of Edinburgh Edinburgh, UK

Fall 2017

Lab Demonstrator · Marker Robotics: Science and Systems

- Supervised a class of 28 MSc students who designed, built, and programmed autonomous LEGO robots;
- Evaluated the progress of each group throughout the semester and marked their final technical reports.

University of Porto, Porto, Portugal

Fall 2015 and Fall 2014

Teaching Assistant Computer Laboratory

- Taught two MSc classes (~15 students each) OOP in C, Bash scripting, VESA graphics, and state machines;
- Wrote tutorials for developing a game from scratch in Minix (difusal.blogspot.pt/2014/07/minix-posts-index.html);
- Contributed to a ~10% increase in the number of approved students. Boosted quality of the students' final projects.

University of Porto, Porto, Portugal

Spring 2015

Teaching Assistant Computer Graphics

- Taught computer graphics (meshes, visibility, lighting, and shading) to two MSc classes;
- Contributed to the development of *WebCGF*, a WebGL-based library still being used in the course.

PUBLICATIONS

H. Ferrolho, V. Ivan, W. Merkt, I. Havoutis, S. Vijayakumar, “RoLoMa: Robust Loco-Manipulation for Quadruped Robots with Arms,” *Autonomous Robots (AuRo)*, 2023.

C. Mastalli, W. Merkt, J. Marti-Saumell, **H. Ferrolho**, J. Sola, N. Mansard, S. Vijayakumar, “A Direct-Indirect Hybridization Approach to Control-Limited DDP,” *Autonomous Robots (AuRo)*, 2022.

H. Ferrolho, W. Merkt, C. Tiseo and S. Vijayakumar, “Residual Force Polytope: Admissible Task-Space Forces of Dynamic Trajectories,” *Robotics and Autonomous Systems (RAS)*, 2021.

H. Ferrolho, V. Ivan, W. Merkt, I. Havoutis, S. Vijayakumar, “Inverse Dynamics vs. Forward Dynamics in Direct Transcription Formulations for Trajectory Optimization,” in *IEEE International Conference on Robotics and Automation (ICRA)*, Xi’an, China, 2021.

H. Ferrolho, W. Merkt, V. Ivan, W. Wolfslag, S. Vijayakumar, “Optimizing Dynamic Trajectories for Robustness to Disturbances Using Polytopic Projections,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Las Vegas, USA, 2020.

H. Ferrolho, W. Merkt, Y. Yang, V. Ivan and S. Vijayakumar, “Whole-Body End-Pose Planning for Legged Robots on Inclined Support Surfaces in Complex Environments,” in *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, Beijing, China, 2018.

Y. Yang, W. Merkt, **H. Ferrolho**, V. Ivan and S. Vijayakumar, “Efficient Humanoid Motion Planning on Uneven Terrain Using Paired Forward-Inverse Dynamic Reachability Maps,” *IEEE Robotics and Automation Letters (RA-L)*, 2017.

🧑 INVITED TALKS, WORKSHOPS, AND LECTURES

Advanced Research + Invention Agency (ARIA) · Cambridge, UK Nov 2025

Judge Vsim Hackathon: How to Train Your Robot 🌐 v-sim.co.uk

IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS) · Hangzhou International Expo Center Oct 2025

Workshop Organizer The Art of Robustness: Surviving Failures in Robotics 🌐 art-of-robustness-iros2025.github.io

Georgia Institute of Technology · AI4OPT PhD Course: Optimal Control and Learning Sep 2025

Seminar Trajectory Optimization with Direct Transcription and Whole-Body Robot Dynamics 🌐 Course Materials

ROSCon UK · Edinburgh, UK Sep 2025

Programme Committee member 🌐 roscon.org.uk/2025/

Beijing Institute for General Artificial Intelligence (BIGAI) · Beijing, China Feb 2025

Talk Robotics, Optimal Control, Trajectory Optimisation, Julia Robotics 🌐 www.bigai.ai

Delft University of Technology · Robotics: Science and Systems (RSS) Conference Jul 2024

Workshop Organizer Tactile Sensing for General Purpose Robot Learning 🌐 noosphereworkshop.github.io

Ocado Technology · 10th Research Conference (Internal) Apr 2024

Talk Is the robot grasping something? Binary Grasp Classification of RGB Frames

Session Chair Robotics Simulation

Massachusetts Institute of Technology (MIT) · JuliaCon 2023 Jul 2023

Talk Using Julia to Optimise Trajectories for Robots with Legs 📺youtu.be/5uF3VqgjiVE

Heriot-Watt University · CDT RAS Annual Conference Nov 2022

Short Talk (Alumni Panel) My journey as a Robotics CDT student

University of Porto · Intelligent Systems, Interaction and Multimedia Seminar Dec 2020

Talk Trajectory Optimization with Direct Transcription

University of Edinburgh · Institute of Perception, Action and Behaviour May 2019

Seminar Robust Dynamic Trajectory Optimization

ENEI · Portuguese National Meeting of Informatics Students Mar 2018

Workshop Introduction to Robotics with Three.js

University of Porto · Computer Laboratory Dec 2014, Dec 2015

Lecture (MSc) Graphics, State Machines, and OOP in C for MINIX · bit.ly/lcom-ferrolho-2014, bit.ly/lcom-ferrolho-2015

♡ VOLUNTEERING / ⓘ OTHER INFORMATION

ELLIS PhD & PostDoc Program, Evaluator for the pre-screening of PhD applicants (European Union) 2020

Edinburgh International Science Festival, Educator at the drop-in science event (Edinburgh, UK) 2018

IEEE RAS International Conference on Humanoid Robots, Organizing Staff (Birmingham, UK) 2017

Engineering and Physical Sciences Research Council, PhD scholarship (Edinburgh, UK) 2017

Microsoft, Microsoft Student Partner (Porto, Portugal) 2016

Conservatory of Music, Piano lessons (Viseu, Portugal) 2004 – 2009

Fun Languages, English studies (Viseu, Portugal) 2002 – 2010